TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Neps of Raw Cotton and Sliver

Counting the small tangled knots that survive spinning to become thick places in yarn and white specks in dyed fabric — in the raw cotton, and again in the card sliver to prove the card has done its work.

TE-3104 · QUALITY EVALUATION OF FIBRE AND YARN SESSIONAL · EXPERIMENT 4 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to determine the nep content of raw cotton and of card sliver, by the AFIS counting method (neps per gram, with nep size and nep type) and by the manual card-web counting method (neps per 100 square inches of web), and to calculate the nep removal efficiency achieved between raw cotton and sliver.

1.1 What a nep is

A **nep** is a small, tight entanglement of fibres — a knot typically 0.5–1.5 mm across, far smaller than a yarn diameter is long, but large enough to disturb drafting, appear as a thick place in yarn, and refuse dye in fabric. Neps are both born and killed inside the mill: they arrive with the bale, they are created by harsh mechanical action on immature fibres, and they are removed — mainly at the card — by the intense fibre-by-fibre opening between cylinder and flats.

Two types must be distinguished because they have different causes and different cures:

- **Fibre neps (biological and mechanical neps of fibre alone)** — tangles of fibres with no foreign core, formed when thin-walled, immature fibres collapse and roll together during ginning, opening and carding. Their number is a fingerprint of maturity (Experiment 1) and of how gently the fibre has been handled.
- **Seed-coat neps** — a fragment of broken seed coat with fibres still attached, torn from the seed during aggressive ginning. The dark fragment shows in yarn and fabric as a coloured speck that no amount of carding fully disperses; prevention is at the gin, by not breaking the seed.

1.2 Why two samples: raw cotton and sliver

A nep count on raw cotton alone is a property of the bale. A nep count on the card sliver made from that cotton is a property of the *process*. The difference between them — the nep removal efficiency — is the standard audit of blowroom and card performance:

$$\text{Nep removal (\%)} = (\text{neps/g in raw cotton} - \text{neps/g in sliver}) / \text{neps/g in raw cotton} \times 100$$

A well-set card on a suitable mixing removes roughly 75–90% of the incoming neps while generating few new ones. If sliver neps are high, the fault may be incoming (immature, nep-heavy cotton), mechanical (blunt or overloaded card wire, wide settings, excessive production rate), or both — and the raw-cotton count taken in this experiment is what allows the two causes to be separated.

1.3 The two counting methods

AFIS method. The sample is individualised as in Experiment 3; the sensor recognises a nep by its characteristic signal — a sudden, dense, short event quite different from a straight fibre — and reports neps per gram, mean nep size, and the split between fibre neps and seed-coat neps. It is fast, objective and size-sensitive, and it is the reference method for process studies.

Manual card-web method. The classical mill method needs no electronics: a known area of card web is taken on a black velvet board or counted in place under standard lighting against a counting template, and every visible nep in the area is tallied with a hand counter. Results are reported as neps per 100 in² of web:

$$\text{Neps per 100 in}^2 = \text{neps counted} \div \text{area counted (in}^2) \times 100$$

The web method counts only neps large enough to see (roughly the same population that will show in yarn), which is its practical virtue; its weakness is operator dependence, which is controlled by standard illumination, a fixed minimum nep size judged against reference photographs, and duplicate counts by two operators.

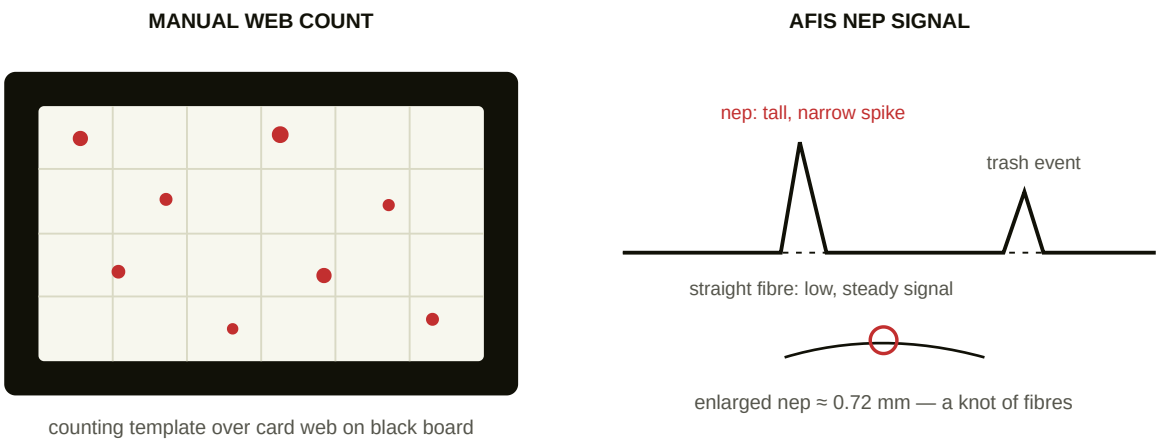


Figure 1 — Left: the manual counting template laid over card web on a black board; red marks stand for the neps tallied in each grid square. Right: how the AFIS sensor sees a nep — a tall, narrow density spike against the low, steady signal of a straight fibre.

2. Apparatus and procedure

- **Apparatus:** AFIS with nep module; black velvet boards, counting template (known area, e.g. 10 in × 10 in) and hand tally counter for the web method; standard counting illumination; card (for web and sliver sampling); conditioned samples of raw cotton from the identified bale or laydown and of the sliver produced from it.
- **Sampling:** the raw cotton and the sliver must come from the *same* laydown, or the removal calculation is meaningless. Take raw cotton from at least five points of the mixing; take sliver from a full can produced under steady card conditions, discarding the first length after any setting change.
- **AFIS count, raw cotton:** run at least three 0.5 g replicates; record total neps/g, fibre neps/g, seed-coat neps/g and mean nep size in micrometres. Condition the sample first — damp cotton individualises poorly and under-counts.
- **AFIS count, sliver:** take sliver lengths totalling the replicate mass without drafting or stretching them, feed gently, and run the same number of replicates on the same settings.
- **Manual web count:** at the card delivery, lay the template over the web (or transfer a web sheet carefully to the black board without stretching it), illuminate at the standard angle, and count every nep meeting the reference minimum size, square by square. Count at least 100 in² in total, in several positions across the web width — neps often concentrate near one edge where the card is worn.
- **Calculation:** average the replicates, express web results per 100 in², AFIS results per gram, and compute nep removal from the two AFIS means. Report nep type and mean size alongside every count.

3. Worked calculation

AFIS results (model data). Raw cotton: **265 neps/g**, mean nep size 0.72 mm. Card sliver from the same laydown: **42 neps/g**.

$$\text{Nep removal} = (265 - 42) / 265 \times 100 = 223 / 265 \times 100 \approx 84.2\%$$

A removal of about 84% sits squarely in the healthy 75–90% band for a well-maintained card: the blowroom has opened without excessive nep generation, and the card wire and settings are doing their work. The nep population has been cut to about one-sixth of its incoming level before drawing — later processes can only add neps (through drafting and winding abuse), never remove them in bulk, so this card-stage figure largely fixes the yarn's nep fate.

Mean nep size. At 0.72 mm the average nep is of the size the card is best at removing and the eye is just able to see in web. A rising mean size alongside a rising count points to seed-coat fragments and harsh ginning; a falling mean size with a rising count of very small neps points to immature-fibre nep generation inside the mill — the two patterns demand opposite remedies, which is why size is always reported with count.

Manual web count (illustrative tally). Counting 100 in² of the same card's web in ten template positions gives tallies of 2–5 neps per 10 in² square, totalling 34 neps:

$$\text{Web neps} = 34 / 100 \text{ in}^2 = 34 \text{ neps per } 100 \text{ in}^2$$

For card web, single-figure counts per 100 in² are excellent and counts in the 30s are a common working level for medium-quality mixings; a sudden doubling against the mill's own control chart — not any universal table — is the signal to inspect wire condition, flat settings and production rate. The web figure and the sliver AFIS figure are different statistics of the same web (visible neps per area vs all neps per gram) and should trend together; if they diverge, suspect the sampling before suspecting the card.

> **DIAGNOSIS TABLE**
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> High raw-cotton neps + high sliver neps + normal removal % → an incoming-material problem (immature or gin-damaged cotton): change the mixing. Normal raw neps + high sliver neps + low removal % → a machine problem: inspect card wire, settings and rate. Rising seed-coat share → a ginning problem: no in-mill setting cures it fully.

4. Precautions

- Match the samples: sliver must be traced to the same laydown as the raw cotton, sampled after the card has settled at steady conditions. Comparing Tuesday's sliver with Monday's bale produces a

removal figure for a process that never existed.

- Do not stretch, draft or compress sliver while sampling — handling a sliver can both break neps apart and roll loose fibres into new ones.
- Condition raw cotton before AFIS testing and run the full set of replicates; nep counts are inherently scattered, and a single 0.5 g replicate can miss the mean by 20%.
- In the web count, hold the minimum visible nep size constant against the reference standard and use the same lighting for every count; a brighter lamp on Friday must not be allowed to "find" a nep problem that is really a counting problem.
- Count across the full web width. Edge-concentrated neps are diagnostic of local card wear, and a centre-only count hides exactly that fault.
- Record nep type. Reporting one merged number discards the distinction between a maturity problem the mill can blend away and a seed-coat problem it cannot.

Practice (viva preparation)

1. Define a nep, and distinguish a fibre nep from a seed-coat nep by origin and by cure.
2. Raw cotton carries 265 neps/g and its sliver 42 neps/g. Calculate the nep removal and judge the card's performance.
3. Why can later processes not rescue a nep-heavy sliver?
4. What does the mean nep size add to the information in a nep count? Give the two contrasting diagnoses.
5. Why does immature cotton form fibre neps so readily?
6. State two weaknesses of the manual web count and the controls used against each.

Answer key

1. A nep is a small tight entanglement of fibres. A fibre nep is a knot of fibres alone, formed from collapsed immature fibres during ginning and mechanical processing; it is reduced by blending maturity, gentle opening and good carding. A seed-coat nep is a fragment of seed coat torn off at ginning with fibres attached; it can only be prevented at the gin, and in the mill it can be partly removed but never fully dispersed.
2. Removal = (265 − 42)/265 × 100 ≈ 84.2% — inside the healthy 75–90% band, so the blowroom and card are performing well on this mixing; the residual 42 neps/g is a normal working level for sliver.
3. Nep removal requires the intense, fibre-by-fibre opening action that exists only at the card (and, in combing, at the comber, which is a separate preparation route). Drawing, roving and spinning only draft and twist: they may break a nep's fibres or bury it, but they do not open knots out — and careless drafting creates new ones.

4. Size separates causes. Large mean size with seed-coat content points to gin damage — broken seed — which mill settings cannot cure. A population of many small neps points to immature fibres rolling up during opening and carding — a maturity and handling problem that blending and gentler settings can improve.
5. An immature fibre has a thin secondary wall and a wide lumen, so it is a flat, limp ribbon with almost no torsional or bending stiffness. Any rubbing action — gin saws, opening beaters, card wire — rolls such ribbons into balls instead of straightening them, and once rolled they entangle neighbouring fibres into a stable knot.
6. Operator dependence in judging what counts as a nep, controlled by a fixed reference minimum size, standard lighting and duplicate counts; and sampling bias across the web width, controlled by counting template positions spread over the whole width including both edges. (A third accepted weakness: it sees only visible neps, whereas AFIS counts the full population.)

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